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# MAGNETIC STRATIGRAPHY OF THE SIWALIK GROUP, BHITTANI, MARWAT AND KHASOR RANGES, NORTHWESTERN PAKISTAN AND THE TIMING OF NEOGENE TECTONICS OF THE TRANS INDUS

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Abstract. Nine stratigraphic sections in the Siwalik Group, three from the Bhittani Range, five from the Marwat Range, and one from the Khasor Range, were measured and sampled for magnetostratigraphic studies. Progressive alternating field and partial thermal demagnetization of samples indicated the presence of two components of magnetization. These are a secondary component of low coercivity, which can be removed by alternating field demagnetization in fields up to 40 mT and/or temperatures to 400°6, and a stable primary component of magnetization which has blocking temperatures above 450°C. Therefore all samples were subjected to blanket demagnetization at 450°C and 500°C, and some to temperatures up to 600°C. Positive fold tests from the Bhittani and Marwat ranges indicate the successful isolation of a prefolding characteristic direction of magnetization. A magnetic polarity reversal sequence for each of the sampled sections is determined and correlated to the magnetic polarity time scale utilizing a fission track date and fauna from the sequences which have previously been shown correlated to the magnetic polarity time scale. This correlation shows that the Siwalik Group in the area of study ranges in age from the late Gilbert chron to the early Brunhes chron (about 0.5 m.y. B.P.). Deposition of the thick sandstone units of the Siwalik Group in the Marwat Range began at least by Gauss chron time. The correlation of observed magnetic polarity stratigraphy, of the Khassor Range, with the standard magnetic polarity time scale shows that the Siwalik Group of this area ranges in age from the early Gauss chron (about 3.2 Ma) to the Brunhes chron. The region of this study began to subside prior to 4 Ma and subsided at a rate of from 60 to 100 cm per 1000 years. The tectonism of the region began within the last 1 m.y. and may well be synchronous with folding in the Pabi Hill anticline on the other limb of the syntaxis near Jelum. This orogeny is therefore one of the world's youngest and is undoubtedly still in progress.

## 1. Introduction

In 1963, Cox et al. [1963] wrote an epochal paper creating a dated polarity stratigraphy. This study and others which followed it opened a new era in tectonics and stratigraphy. The paper presented here utilizes the magnetic time scale pioneered by those authors and is similar to a host of others that have revolutionized the geochronology of the Mesozoic and Cenozoic. It is with great sadness that we submit this paper to a volume commemorating the late Allan Cox. His impact on earth science is profound and will not be forgotten. His incisive comments at scientific meetings will be sorely missed as well as his friendship.

The fluviatile Neogene Siwalik Group rocks have attracted the attention of various workers for well over a century [Meddlicot, 1876; Wynne, 1877] because of the picturesque exposures, easy accessibility, and wealth of vertebrate fossils. Type localities of the various formations of the Siwalik Group are located in the Potwar Plateau of Pakistan. Since the publication of Wynne's [1880] memoirs the Siwalik sediments of the Trans-Indus Salt Range (Figure 1) have attracted the attention of several workers; however, work in this area was difficult because of logistics and frequent tribal clashes. The Siwalik Group on both sides of the Indus River was considered to have the same age and origin because of the similarity of the lithological facies. This resulted in usage of Potwar Plateau Siwalik Group nomenclature for the upper Cenozoic sediments of the Trans-Indus Salt Range [Hemphill and Kidwai, 1975; Fatmi, 1974] Morris [1938] recognized that the sediments of the Marwat and Bhittani ranges were much younger in general than those of the Potwar Plateau, since they contained a Plio-Pleistocene fauna. He therefore proposed a new nomenclature of the Siwalik sediments of this area. Recent work by Munthe [1981] and Khan and Opdyke [1981] has confirmed the Plio-Pleistocene age for the Siwalik sediments in the vicinity of the Sheikh Budin Hills and has emphasized the need to accept Morris's nomenclature for the Siwaliks of the Trans-Indus area. This stratigraphic nomenclature has been slightly altered by Munthe [1981] and his usage is followed in this report.

The units are, from older to younger, the Kargocha, Marwat, and Malagan formations. These formations are exposed in the Bhittani, Marwat, and Khassor ranges (Figure 1) and constitute the southwestern extension of the Trans-Indus Salt Range. The western portion of the Bhittani Range lies in tribal territory and is therefore inaccessible. The part of the Bhittani Range studied along the Bain Pass highway and east of the Bain Pass consists of an asymmetric anticline with a major fault along the fold axis. Three sections were measured and sampled for paleomagnetic studies. These were Bain Pass "A", which essentially follows the main road through the Bain Pass, Bain Pass "B", which is a replication of the lower part of the section taken about 1 km southeast of the road. The third section is at Garhi Landa (measured section 2 in Figure 1) on the southern flank of the anticline.

The Marwat and Khassor ranges are two parallel asymmetric anticlines whose axes generally trend NE-SW, and extend between the Kurram River and the southeastern part of the Bhittani Range (Figure 1). The Marwat Range is the northernmost anticline and consists of Siwalik sediments which are almost overturned at its northern termination near the Kurram River. The Khasor Range is the southernmost anticline, whose core consists of Paleozoic and Mesozoic sedimentary rocks. The Siwalik Group unconformably overlies the Mesozoic rocks and forms the outer part of the anticline. The Siwalik sediments are exposed only along the northwestern flank of the fold.

## 2. The Siwalik Group

The Siwalik Group of the Bhittani Range consists of alternating sandstone and siltstone/claystone units of varying thickness, compactness, and colors. The sandstones are brown and compact in

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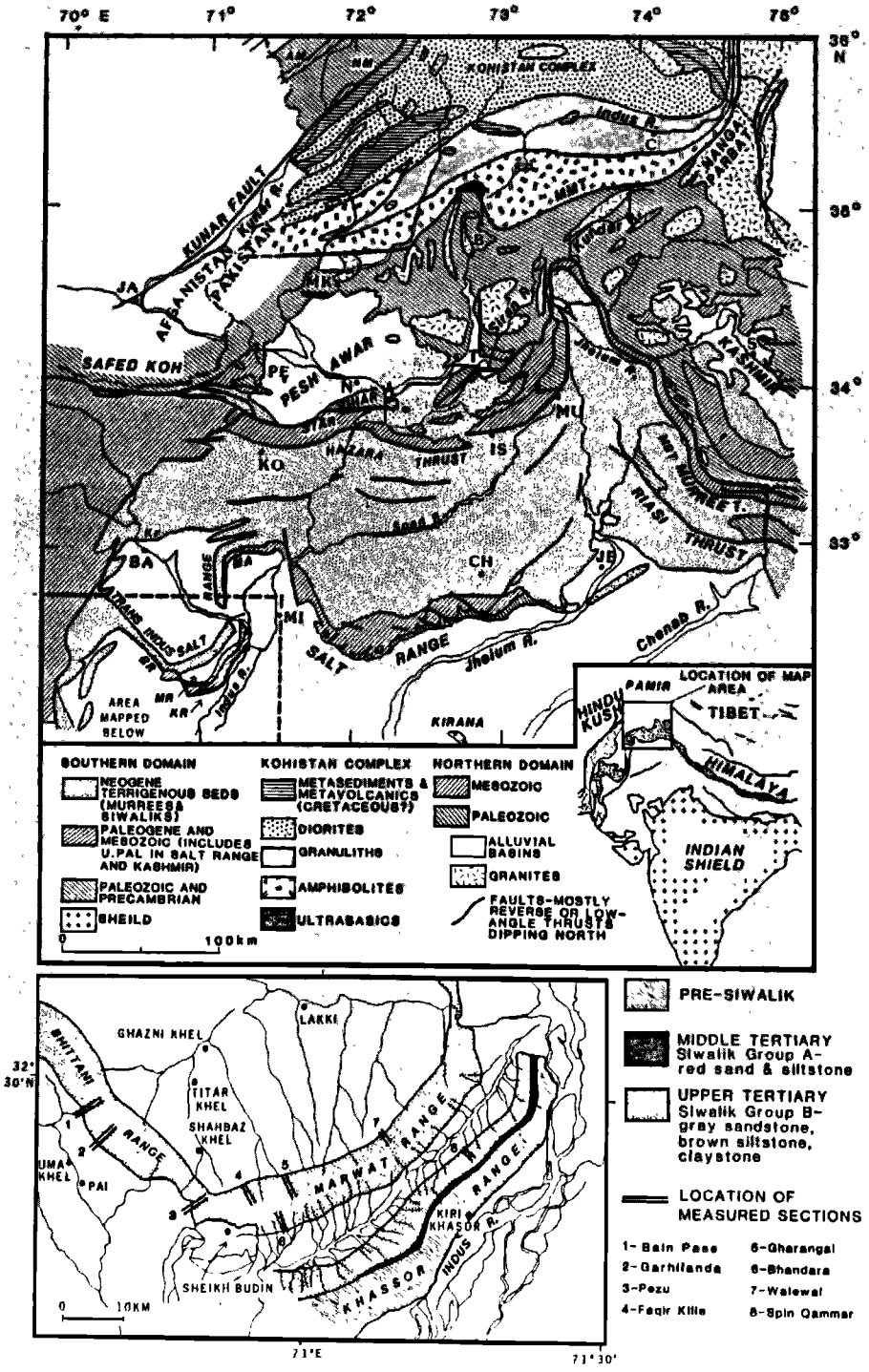



Fig. 1. Geologic sketch-map of northern Pakistan.

the middle and upper parts of the stratigraphic sequence. The siltstones and claystones are bright red at the base, whereas yellowish gray, pale orange, and pale brown colors predominate in the middle and upper parts of the sequence. A few gray colored siltstones and claystones are also present in the middle part of the stratigraphic sequence. Thicknesses of sandstone units vary from less than 4 m to more than 50 m, whereas siltstone units vary from 5 to 150 m. At the Bain Pass B section, a distinct 20- to 30-cm-thick bentonitic ash layer is observed 730 m from the base of the stratigraphic sequence. In the same section three other ash layers, each 5 to 10 cm thick, and separated by a few meters of siltstone, are observed at about 810 m from the base of the section. In the Garhi Landa section, an ash layer 5 to 15 cm thick is located 270 m from the base of the sequence. A very distinct tuff unit, 1.5 m thick, occurs 80 m above this unit, and a distinct diamictite unit is situated 625 m from the base of the section. It extends throughout the anticline, easily traced on aerial photographs and was first observed by Morris [1938], who referred to it as the Bain Boulder Bed. This unit is described in detail by Khan et al. [1985].

The Siwalik sediments exposed throughout the Marwat Range consist of a lower part composed of thick, massive, light colored

TABLE 1. General Features of the Various Stratigraphic Sections

| Sect | Stratigraphic Thickness | Orientation of Strata Strike | Dip | Total Sites | Class A Sites | Class B Sites | Class C Sites |
|---|---|---|---|---|---|---|---|
| BP A | 2200 | NW-SE | 45'NE | 46 | 39 (85%) | 7 (15%) | - |
| BP B | 900 | NE-SE | 30'NE | 31 | 20 (65%) | 9 (29%) | 2 (6%) |
| GL | 2200 | NW-SE | 45'SW | 56 | 50 (89%) | 4 (7%) | 2 (4%) |
| PZ | 1000 | NW-SE | 25'NE | 31 | 25 (81%) | 4 (16%) | 1 (3%) |
| FK | 1300 | NE-SW | 25'NW | 35 | 31 (89%) | 4 (11%) | - |
| GH | 1500 | NE-SW | 25'NW | 23 | 21 (91%) | 2 (9%) | - |
| WL | 1000 | NE-SW | 34'NW | 8 | 8 (100%) | - | - |
| BH | 2000 | NE-SW | 55'SE | 21 | 20 (95%) | 1 (5%) | - |
| SQ | 1300 | NE-SW | 30'SW | 21 | 18 (86%) | 2 (14%) | - |

BPA, Bain Pass A; BPB, Bain Pass B; GL, Ghardi Landa; PZ, Pezu Pass, FK, Fakir Kille; GH, Gharangai; WL, Walwai; BH, Bhandara; SQ, Spin Qammar.

sandstones, whereas the upper part consists of an alternating sequence of light gray sandstones and pale brown to pale yellowish gray siltstones. As the Siwalik Group is traced northeastward, the number and thickness of the finer-grained siltstone/claystone units gradually decrease.

The Siwalik Group of the Khassor Range rests disconformably upon the underlying Mianwali Formation of Triassic age. The Siwalik sequence consists of a lower red clay zone and the upper light gray sandstones. The middle and upper part of the Siwalik Group display only a few siltstone units, each 2 to 5 m thick. The red clay zone is not present at the southwestern end of the Khasor Range, where the entire Siwalik sequence consists of a thick sandstone overlying a conglomeratic bed.

## 3. Paleomagnetic Studies

Three to five oriented hand samples were collected from each site using the techniques of Johnson et al. [1975]. The samples were taken from the fine- grained deposits of siltstone and claystone. Sandstones were not sampled since they often contain multidomain magnetite particles which are magentically unstable. In the Bain Pass A section, 46 sites were sampled from a more than 2200-m-thick stratigraphic sequence. The Bain Pass B section is composed of 25 sites sampled from a stratigraphic sequence 900 m thick. Six sites from the Bain Pass A section were also included in this analysis because of overlaps between the sections. The correlation of these two sections was made by tracing a prominent sandstone laterally along strike. Fifty-six sites were sample through 2200 m of the Garhi Landa section. The strata generally strike NW-SE, and the dip is 25° to 65° NE in the case of Bain Pass A and B sections and SW in the case of the Garhi Landa section.

Five sections were measured and sampled from the Marwat Range: the localities are Pesu Pass, Faqir Kille, Gharangai, Walewai, and Bhandara. Exposures are poor along the front of the Khasor Range, and it was possible to sample only one section at Spin Qammar. The locations of all sections from the Marwat and Khassor ranges are shown in Figure 1. The number of sites sampled and the stratigraphic thickness measured and sampled are given in Table 1.

The intensity and direction of remanant magnetization were measured using a Digico spinner magnetometer developed by Molyneaux [1971]. The intensity of natural remnant magnetization (NRM) of samples, collected from all of the sections, ranged from 1 x $10^{-2}$ to 5 x $10^{-2}$ A/m.

### 3.1. Alternating Field Demagnetization

After measuring the NRM, samples were selected for progressive alternating field (AF) demagnetization in steps of 5 mT, up to 80 mT (Figure 2). The results of AF demagnetization suggest that the NRM of most of the samples consists of two components of magnetization: low-coercivity component which is removed by AF treatment of up to 40 mT, and a high coercivity component which is not much affected even after treatment to 80 mT. The low-coercivity component is interpreted as a secondary component of magnetization acquired in the Earth's present-day magnetic field. The high-coercivity component does not show significant change in direction or loss of intensity in fields above 25 mT. The more highly coercive rocks are red or brown in color, clearly indicating the presence of hematite.

Table 2. Directions of Remanent Magnetization

| Locality | N | D° | I° | K | α95 |
|---|---|---|---|---|---|
| Bain Pass A | 19 | 1.0 | 33.8 | 10.0 | 11.1 |
| | 27 | 177.7 | -33.0 | 13.2 | 7.9 |
| Bain Pass B | 16 | 347.9 | 30.2 | 10.5 | 11.9 |
| | 15 | 182.9 | -17.8 | 10.9 | 12.1 |
| Garhi Landa | 21 | 349.3 | 36.9 | 11.8 | 9.6 |
| | 35 | 178.8 | -28.2 | 10.1 | 8.0 |
| Pezu Pass | 14 | 6.5 | 27.9 | 7.5 | 15.4 |
| | 17 | 171.4 | -43.1 | 10.2 | 11.6 |
| Fagir Kille | 10 | 355.4 | 26.9 | 10.3 | 15.7 |
| | 25 | 174.9 | -37.7 | 13.6 | 8.1 |
| Gharangai | 10 | 355.7 | 31.4 | 21.9 | 10.5 |
| | 13 | 176.9 | -36.4 | 15.2 | 10.9 |
| Bhandara | 9 | 7.3 | 49.4 | 20.0 | 11.7 |
| | 12 | 167.4 | 39.4 | 15.4 | 11.3 |
| Walewal | 3 | 350.0 | 28.7 | 18.0 | 29.9 |
| | 5 | 175.2 | -31.2 | 15.1 | 20.2 |
| Spin Qammar | 10 | 347.6 | 40.9 | 29.9 | 8.9 |
| | 10 | 169.9 | -53.9 | 9.9 | 16.1 |
| All Normal | 9 | 355.5 | 34.2 | 70.9 | 6.2 |
| | 9 | 174.8 | -33.6 | 89.0 | 5.5 |

N, number of sites; D, declination; I, inclination; K, precision parameter; α95, 95% confidence level [Fisher, 1963]. Data from all sites derived from thermal demagnetization. First row of each entry presents results for normal polarity sites. Second row presents results for reversed polarity sites.

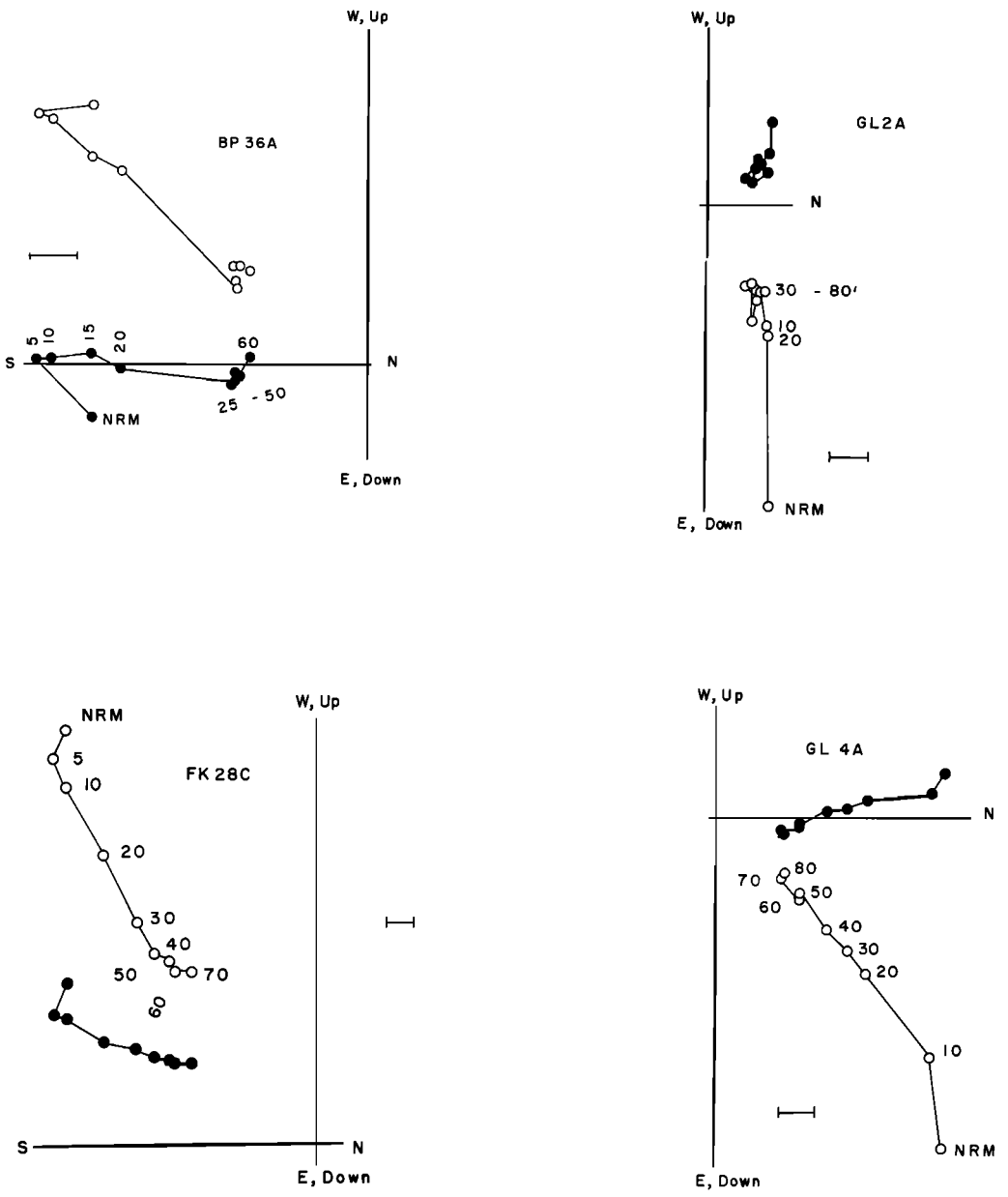



Fig. 2. Typical results of alternating field demagnetization: orthogonal plots based on successive endpoints of AF demagnetization vectors. Solid (open) circles are plotted on a horizontal (vertical) plane. Short bars indicate the scale for remanent magnetization intensity of 10 A/m. The demagnetizing field is in milliteslas. All of these plots are based on bedding-corrected directions of magnetizations.

### 3.2. Thermal Demagnetization

Selected samples were subjected to progressive thermal demagnetization (PTD) at temperatures from 100°C to 660°C, with intervals of 25°C to 100°C, and orthogonal plots of representative results are shown in Figure 3. These results indicate the presence of two components of magnetization in most of the samples. These consist of a low blocking temperature component carried by magnetic grains with unblocking temperatures ranging from 100°C to 400°C, and a high blocking temperature component with blocking temperatures ranging from 450°C to 660°C. The directions and intensity of magnetization often change dramatically up to temperatures of 300°C to 400°C, whereas at higher temperatures, the vectors exhibit a linear decay toward the origin (Figure 3). In some of the samples, the intensity of remanent magnetization drops to only a few percent of the NRM intensity at 575°C. However, some of the samples retain a few percent of the NRM intensity even after heating at 660°C. These studies show that a secondary component of magnetization is present in most samples which is of normal polarity and was probably acquired in the present-day magnetic field of the Earth. This component has unblocking temperatures below 450°C. The characteristic magnetization is revealed at temperatures above 450°C.

All samples from the different sections were subjected to blanket demagnetization at temperatures of 450°C and 500°C. However, some of the samples from pale brown siltstones/claystones, particularly those of the Spin Qammar section, were further demagnetized up to 600°C, so as to be certain of the removal of any secondary component of demagnetization.

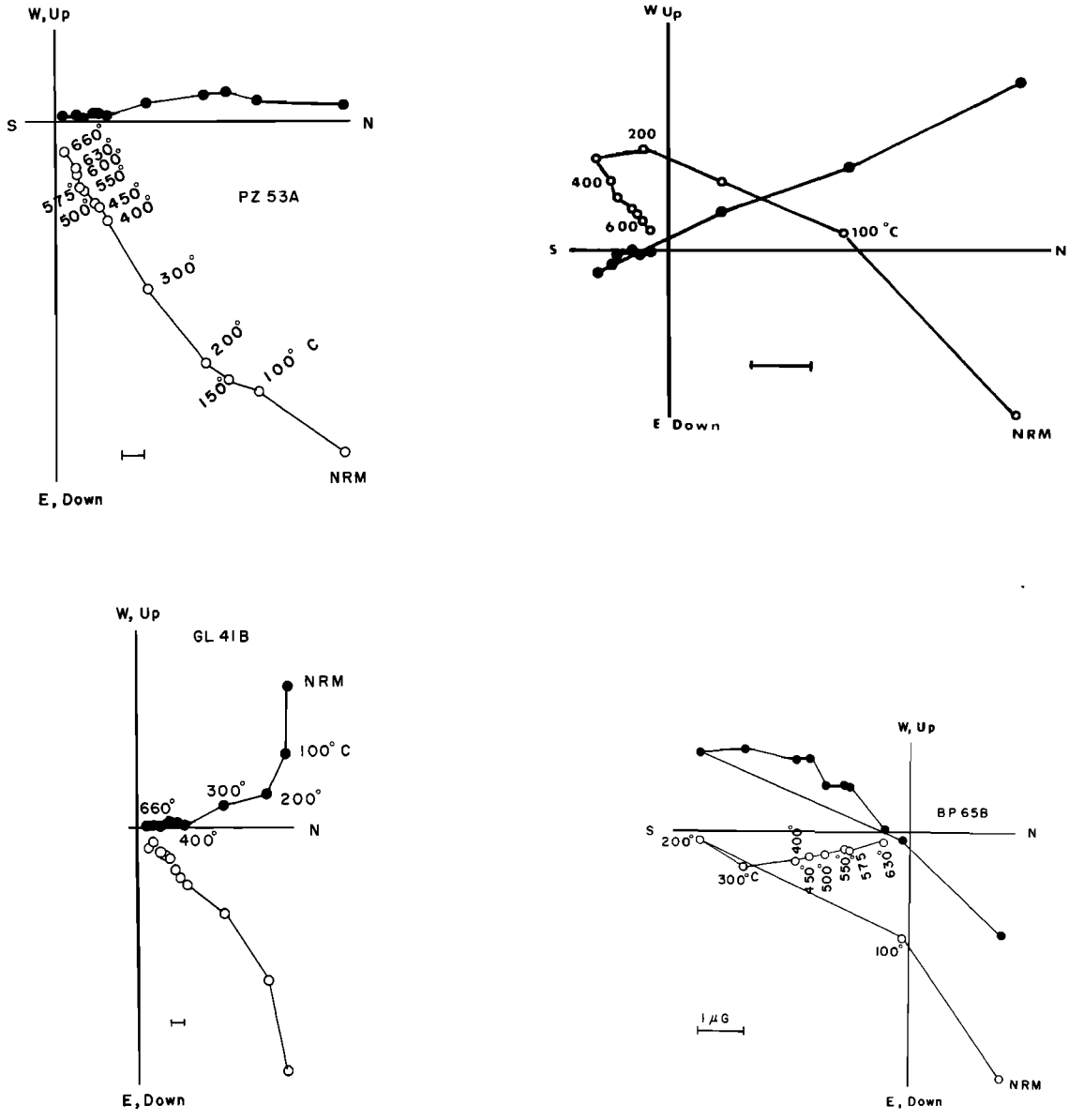



Fig. 3. Results of partial thermal demagnetization. Plotting conventions are same as for Figure 2. Temperatures are in degrees Celsius.

## Tests of Data Reliability

The directions of magnetization obtained at the 500° or 600°C level, after thermal demagnetization, were used to calculate the site-mean directions using the statistics of Fisher [1953]. The site-mean directions which passed the randomness test of Watson [1956] were designated as class A. The sites which failed the randomness test but which had two samples with similar directions and a third deviant direction were designated as class B, whereas the sites with dispersed directions were grouped as class C sites. Only class A and class B type sites were used for further paleomagnetic analysis. The site means were then combined in locality means for both normal and reversed sites (Table 2).

### Reversal Test

Plots of bedding-corrected site-mean directions for both normal and reversed mangetized sites (Figure 4), show that the means of all normal polarity sites (D=355.5, I=34.2) are antipodal to means of all reversed polarity sites (D=174.8, I=33.5) within 1°. This indicates that any overprint acquired in the present field direction is negligible. It is interesting to note that these sediments have a clear inclination error, since the mean direction of magnetization has a significantly lower inclination (34°) than that which can be expected for the dipole field at

TABLE 3. Results of fold test from the Marwat and Bhittani ranges.

| | N | K1 | K2 | K2/K1 | Significant at 99% Level |
|---|---|---|---|---|---|
| Marwat | 44 | 3.5 | 15.0 | 4.29 | Yes |
| Bhittani | 112 | 4.8 | 10.8 | 2.25 | Yes |

K1 (K2) is the value before (after) correcting for folding.

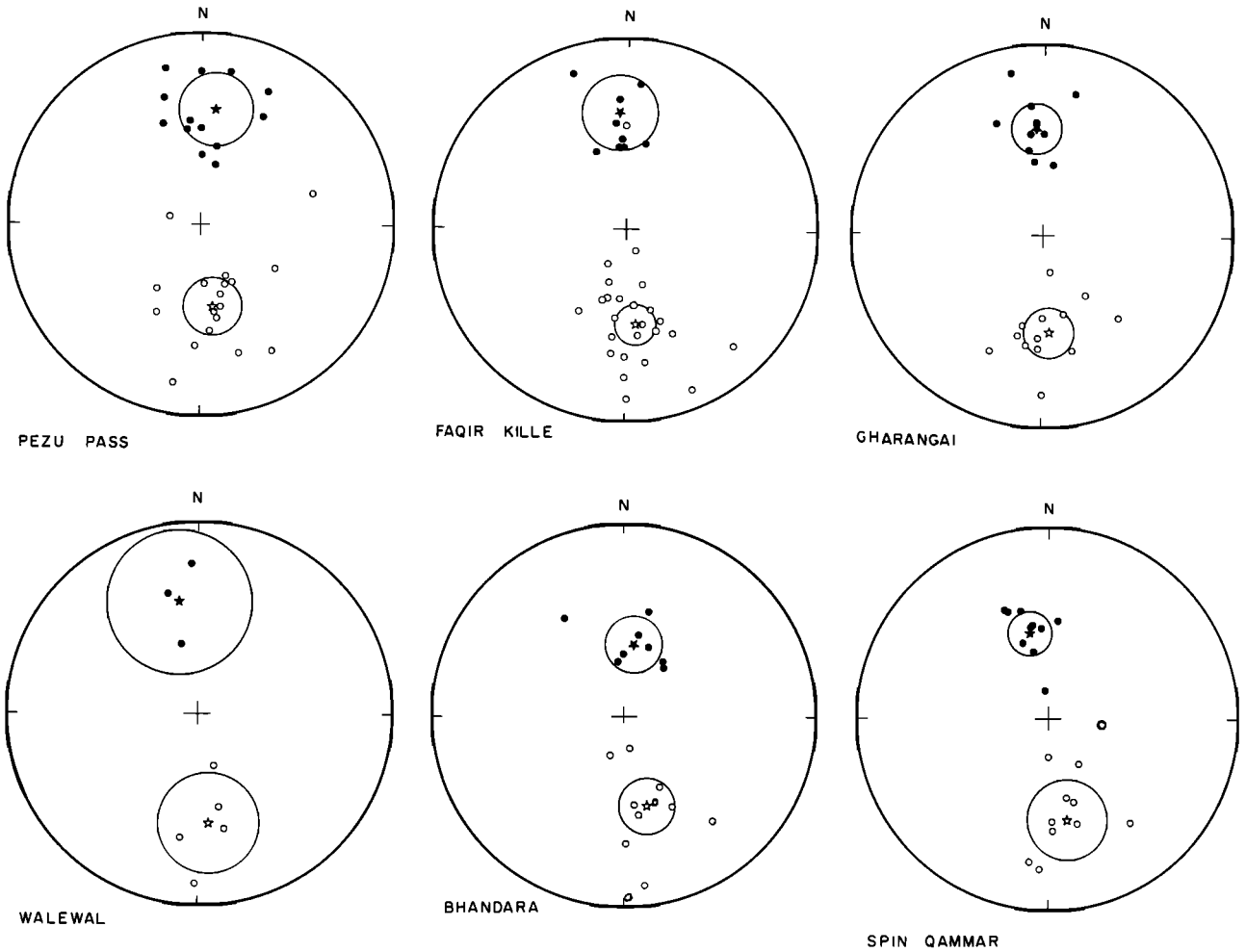



Fig. 4. Equal-angle stereographic projection of bedding-corrected site-mean directions obtained after thermal demagnetization at temperatures up to 600°C. Solid (open) circles are the plots on the lower (upper) hemisphere. Solid (open) stars are the mean of normal (reversed) polarity sites, and large circles represent the limits of 95% confidence level.

this latitude (50°), a situation similar to that reported by Tauxe et al. [1980] for Miocene sediments from the Potwar Plateau.

Fold Test

Sections have been sampled from both flanks of the Bhittanni and Marwat range anticlines, making it possible to perform a fold test for the sediments from each structure. Antipodes of reversed polarity site-mean directions were averaged with site-mean directions from normal polarity sites, and the Fisher [1953] precision parameter was calculated both before and after bedding correction. In each case, the fold test is passed at the 99% confidence level [McElhinny, 1964]. Details are presented in Table 3 and illustrated in Figure 5. This fold test demonstrates conclusively that the magnetization isolated by thermal demagnetization is prefolding, since the sedimentary rocks are of Plio-Pleistocene age and the folding is late Pleistocene; the origin of the magnetization is pre-Brunhes. It is probable that this component was acquired at or soon after deposition, a conclusion that is reinforced by the successful reversal test.

## Magnetic-Polarity Stratigraphy

Virtual geomagnetic pole (VGP) positions were calculated using the thermally demagnetized data corrected for bedding tilt. The north seeking sites (with positive VGP latitude values) were assigned normal magnetic polarities, whereas the south seeking sites (with negative VGP latitude values) were assigned reversed magnetic polarities. The VGP latitudes were plotted against stratigraphic thickness, resulting in a magnetic polarity reversal sequence for each section described as follows, beginning with the Bain Pass A section.

The plot of VGP latitude versus stratigraphic thickness (Figure 6) shows nine magnetozones. All magnetozones are based on more than one site. In order to confirm the validity of the N3 and N5 magnetozones, all samples from these sites were subjected to additional thermal demagnetization at progressively higher temperatures. As no change in direction was observed and since the magnetic properties of these sediments are the same as the sedimentary rocks above and below them, the normal polarity is interpreted as reflecting magnetic field behavior at or near the time of deposition. In addition both of these normally magnetized zones were recollected at a later date in an attempt to confirm the existence and stratigraphic limits of these short normal zones, and careful incremental thermal demagnetization was performed. This additional work served to confirm the existence of these two short magnetozones and helped define their stratigraphic limits.

The methods of analysis used on the Bain Pass section were employed on the other sampled sections. This analysis showed six magnetozones at the Bain Pass B section (Figure 7), seven at Garhi

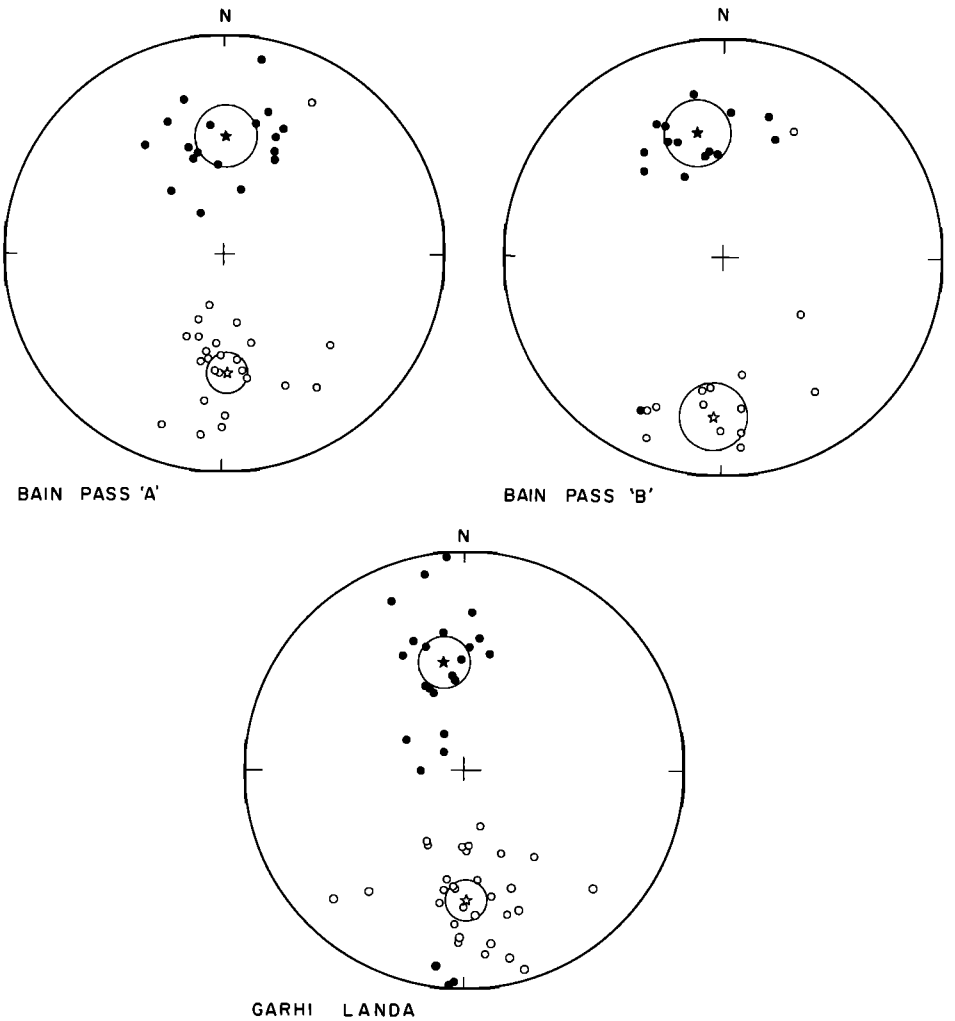



Fig. 4. (continued)

Landi (Figure 8), six at Pezu Pass (Figure 9), and seven at Faqir Kille (Figure 10).

The sections further east become progressively less complete as the amount of massive sandstone in the section increases. The section at Gharangi (Figure 11) for example shows only seven magnetozones; however, sampling in the lower portion of the section is so sparse that it is clear that little can be claimed for the magnetic stratigraphy and many reversals of the field have undoubtedly been missed. The same holds for the Bhandara section (Figure 12), and the Walewal section (Figure 13); the Spin Qammir section is important since it is the only section sampled in the Khassor Range (Figure 14). It is possible to define only three magnetozones from the basal part of the section, which is red in color and has a Kargocha-type facies. Higher in the section it was possible to sample only from four sites, three reversely magnetized and the fourth normally magnetized.

## Correlation of Magnetic Polarity Stratigraphy

Vertebrate fossils collected by Morris [1938] and West [1979], from the Bhittani Range give a Plio-Pleistocene age for the Siwalik sediments exposed in Bain Pass and its vicinity. During the course of paleomagnetic sampling, Stegodon sp. teeth (E. H. Lindsay, personal communication, 1987) were collected from the lower part of the Garhi Landa section. Those teeth are consistent with Plio-Pleistocene age for the Siwalik sediments of the Bhittanni Range [Barry et al., 1982]. Following Opdyke et al. [1979] and Barry et al. [1982], it is most likely that the Siwalik sediments of the Bhittani Range are not older than 4.5 Ma. An ash unit from the base of the section in reversely magnetized sediments in the Bain Pass A section of the Bhittani Range has been dated by Johnson et al. [1982], and yields an age of 2.79 + 0.24 Ma. This date seems to be too old, a situation easily explained by the inclusion of older undetected detrital zircons in the data set. However, it constrains the section to be of Gauss or Matuyama age. Therefore the observed magnetic polarity reversal sequence from three sections of the Bhittani Range are unquestionably Plio-Pleistocene in age and can be unambiguously correlated to the magnetic-polarity time scale of Berggren et al. [1985], as shown in Figure 15.

The normal polarity magnetozone N4 at the top of the Garhi Landa section can be correlated with the Brunhes chron. Dominantly reversed polarity zones from the R1 to R3 magnetozones in the Garhi Landa section and from R2 to the top of the section at N6 magnetozone in the Bain Pass A section are correlated with the Matuyama chron. The long normal polarity magnetozone N2 in the Bain Pass A section, from N2 to N4 magnetozones in the Bain Pass B section, and N1 magnetozone of the Garhi Landa section are correlated with the Gauss chron. The N1 and R1 magnetozones of both Bain Pass sections are correlated with the top of the Gilbert chron. Although they have not been dated, the presence of two ash units at the Gauss/Matuyama boundary in the Bain Pass and Garhi Landa sections supports the validity of this correlation, since dated volcanicity at these times are known throughout the Potwar Plateau [Johnson et al., 1982].

An ambiguity is present in the correlation of the top of the Bain

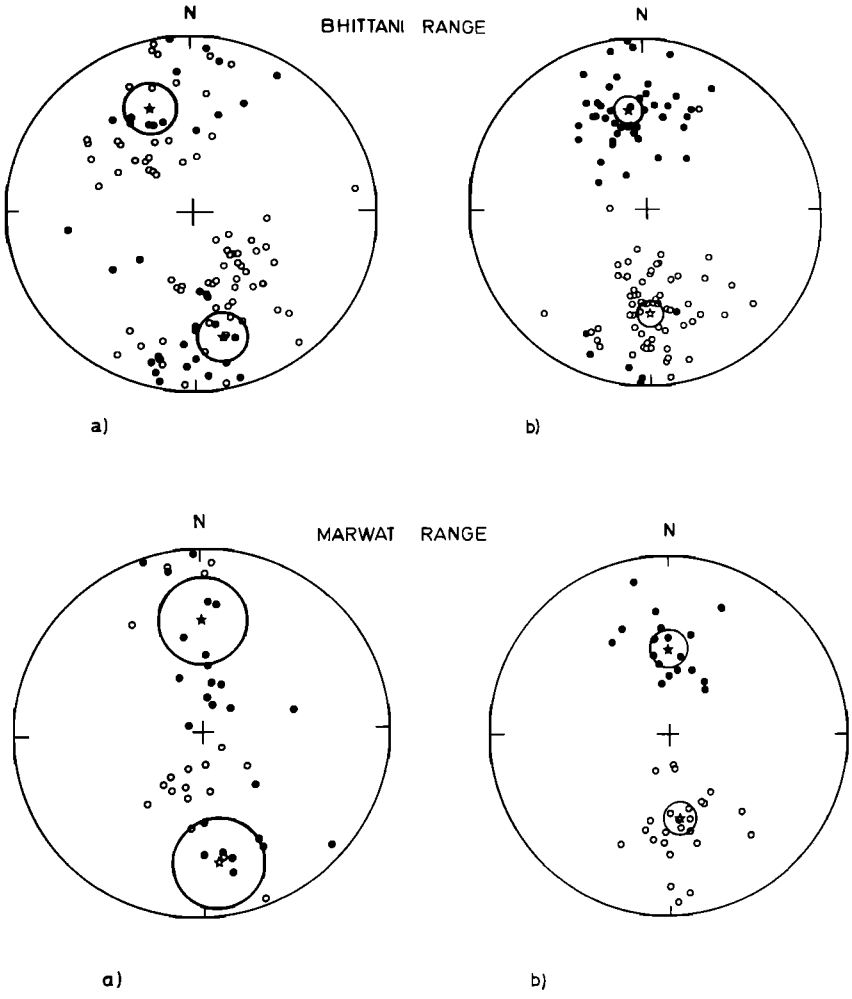



Fig. 5. Plot of site-mean directions after partial thermal demagnetization, from the Bhittani and Marwat range, (a) before bedding correction and (b) after bedding correction.

Pass A section with respect to magnetozone N5. Two possibilities exist: (1) N5 might be the Jaramillo subchron, in which case the top of the section would represent the older portion of the Brunhes chron; or (2) N5 might be the Cob Mountain subchron, in which case the section terminates within the basal portion of the Jaramillo subchron. At present we believe the top of the section is most likely Brunhes in age, since the Malagan Formation is clearly of Brunhes age in other sections, particularly at Garhi Landa; however, the alternative correlation cannot be excluded.

The Garhi Landa section sampled along the southern limb of the fold, covers a time span from the later portion of the Gauss Chron (2.8 Ma) to the Brunhes chron (0.5 Ma). Short-polarity subchrons, like the Reunion, are not observed in this section, probably because of the lack of closely spaced samples. Unfortunately, the Bain Boulder Bed cannot be clearly identified in the Bain Pass. In the Garhi Landa section, however, it is easily identified. Its position in the stratigraphic sequence is shown in Figure 8 and appears to have been deposited at about 2.1 Ma.

Correlation of the Magnetic Polarity Stratigraphy From the Bhittani Range to the Marwat and Khassor Ranges

The correlation between the sections studied in the Marwat Range with those farther west in the Bhittani Range is made difficult by the fact that thick sandstones begin to dominate the sedimentary sections east of Faqir Kille, effectively reducing the sampling intervals, so that the quality of the magnetostratigraphy is seriously degraded. However, the Bain Boulder Bed is present at Pezu Pass stratigraphically below the base of the measured section. This distinctive horizon is believed to be a lahar which was deposited isochronously throughout the region as a result of a volcanic eruption [Khan, et al., 1985] and is therefore an excellent chronostratigraphic horizon. At Garhi Landa, it occurs 100 m below the Olduvai subchron in reversely magnetized sediments. The section at Pezu therefore is entirely younger than this datum. The Bain Boulder Bed is also present in the Faqir Kille, Gharangai, and Bandara sections to the east of Pezu Pass and constrains these sections to be lower Matuyama and younger above this marker.

Therefore it is most likely that magnetozone N2 at Faqir Kille and N3 of the Bhandara and Gharangai sections represent the Olduvai subchron. Correlation of the Walewal section is suspect, and is so indicated by the question marks on the correlation diagram (Figure 16). The correlation is good enough to identify the dominantly reversed polarity magnetozones as the Matuyama chron. The normal magnetozones in the upper part of the Matuyama at Pezu Pass, Faqir Kille, and Garangai are correlated with the Jaramillo subchron. At Pezu and Garangai a second normal event is present below what is believed to be the Jaramillo and it is thought to represent the Cob Mountain subchron. The recovery of sparse Plio-Pleistocene Pinjor age fossils from the region [West, 1979] at least supports the correlation of the dominantly reversed intervals of these sections with the Matuyama chron because, farther to the east in the Potwar Plateau, Pinjor age fossils are known to be upper Gauss or younger [Opdyke et al., 1979].

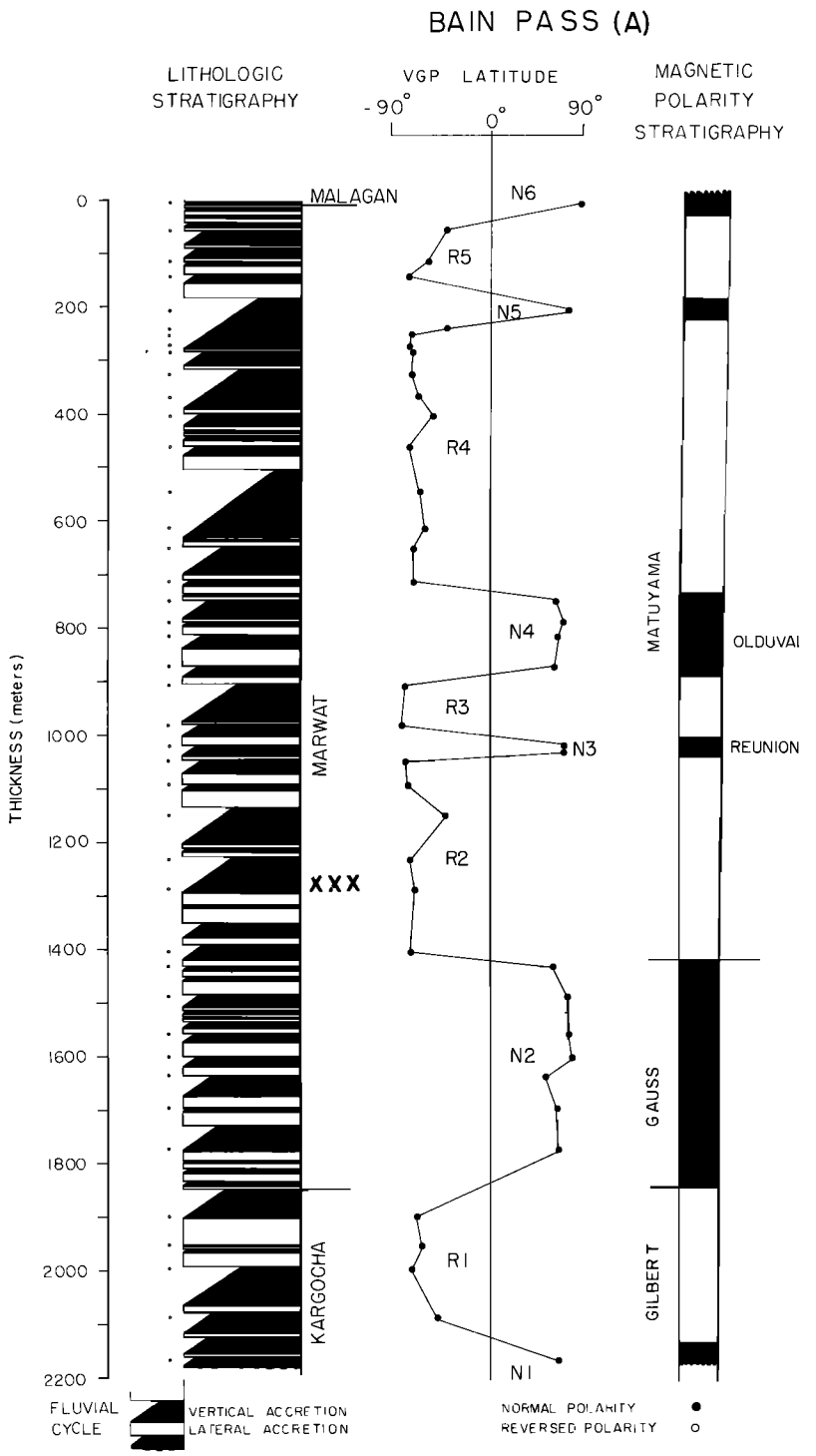



Fig. 6. Plot of VGP latitudes against stratigraphic thickness. The solid (white) blocks in the lithologic stratigraphic column represent vertical accretion-siltstone/ claystone (lateral accretion-sandstone) deposits of the molasse sequence. Solid (white) columns of magnetic polarity stratigraphy represent normal (reverse) magnetic polarities. The boundaries of polarity magnetozone reversals are placed at intermediate position between two successive sites having opposite polarities. The three crosses sign to the right of the lithologic stratigraphic column represent location of ash unit, which has an age of 2.79 + 0.24 m.y. B.P. [Johnson et al., 1982].

The correlation of the lower part of the section at Gharangai and Bhandara is more problematical due to sparse sampling. At Gharangai (Figure 11), for instance, only two sites could be sampled in the lower part of the measured section, and both are normally magnetized. It is possible therefore that the upper site, which lies just below the Bain Boulder Bed, is either within the Reunion subchron or the Gauss chron. An unambiguous determination cannot be made.

At Bhandara, however, even though only five localities could be sampled in the lower 700 m, three of these sites record a polarity change from normal to reversed. This change of polarity most likely

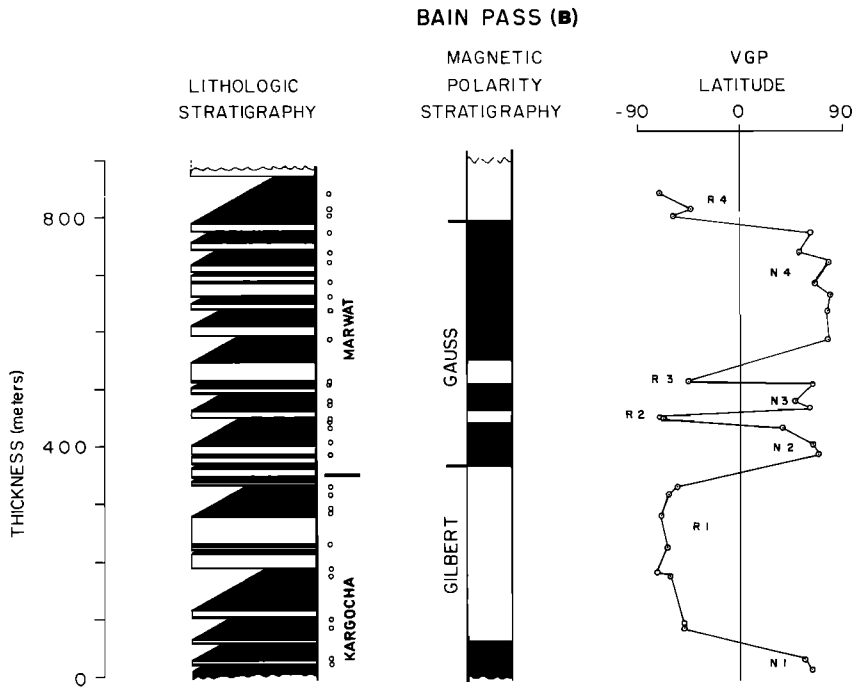



Fig. 7. Bain Pass B. Plot of VGP latitudes versus. stratigraphic thickness. Plotting conventions are same as for Figure 6.

represents the Gauss-Matuyama boundary. The lowermost sites are reversed and normal and most likely represent one of the subchrons within the Gauss and the lower Gauss, respectively (Figure 12).

## Khassor Range

The single section studied in the Khassor Range has only four sites in the upper 1200 m of the section; however, the sampling density in the bottom part of the section is adequate to define three magnetozones (Figure 14). Although it is not possible to be unambiguous, the sequence is best interpreted as the two reversed subchrons, the Kaena and Mammoth of the gauss Chron, and are so interpreted here. Unfortunately, the Bain Boulder Bed is not seen in this section.

## Conclusions

The magnetostratigraphic correlation suggests that the Siwalik Group exposed along the Bain Pass section ranges in age from the Cochiti subchron (3.85 Ma) to the lowermost Brunhes chron (0.70 Ma). During the time between the Gauss/Matuyama boundary (2.48 Ma) and the base of the Brunhes, 17 fluvial cycles of approximately equal thickness were deposited. This suggests that each fluvial cycle was deposited in about 80,000 years. The Garhi Landa section covers a time span from the lower Gauss chron (2.8 Ma) to the top of the Brunhes chron (0.5 Ma). The Bain Boulder Bed lies below the Olduvai subchron and appears to have been deposited at about 2.1 to 2.2 Ma, close in time to the Reunion subchron.

The Siwalik Group rocks of the Pezu Pass section extend from the early Olduvai subchron (about 1.7 Ma) to the top of the Matuyama chron (about 0.75 Ma). The Faqir Kille and Gharangai sections cover the time span between the late Gauss chron (about 2.7 Ma) to the early Brunhes chron (about 0.6 Ma), whereas the Bhandara section is expected to range from the early Gauss chron (3.3 Ma) to the early Brunhes chron. The Walewal section most probably covers the time from the late Olduvai subchron to the early Brunhes chron. Therefore the thick sandstone units of the Marwat Range, which are devoid of siltstone units, were deposited during medial to late Pliocene time.

In the Khasor Range, it is most likely that the Siwalik Group ranges in age from the early Gauss chron (about 3.2 Ma) to the late Matuyama chron (about 0.9 Ma).

In this study, two subchrons which have been difficult to distinguish in magnetostratigraphic studies appear to have been unambiguously identified. These are the Cob Mountain and Reunion subchrons. The Cob Mountain subchron has apparently been identified at Pezu Pass, and in the Gharangai section, it is best expressed at Pezu Pass, where is has been identified in six consecutive sites representing 30 m of the section. Extrapolating the sedimentation rate from the Jaramillo yields an estimated age for the upper and lower boundaries of 1.025 to 1.045. Ma and a duration of 20,000 years. The original date given by Mankinen et al. [1978] was 1.1 m.y., which considering the errors in both methods of dating, must be considered a good agreement.

Magnetozone N3, from the Bain Pass A section, has been correlated to the Reunion subchron. This magnetozone, as interpolated from the base of the Olduvai to the Gauss/Matuyama boundary, yields an age of 2.025 to 2.050 Ma, approximately 25,000 years in duration.

### Rates of Sediment Accumulation

Rates of sediment accumulation were calculated for the sediments of the Marwat Formation in all sections which have reliable magnetic stratigraphies for intervals within the Matuyama chron. The rates of sediment accumulation are higher in the Bhittani Range, with rates of 102 cm/1000 years recorded at Garhi Landa and 81 cm/1000 years at Bain Pass A. The rates of sedimentation at Pezu, for the upper part of the Matuyama chron, yield a rate of 84 cm/1000 years. The sections further east at Faqir Kille (66 cm/1000 years) and Bhandara (65 cm/1000 years) yield rates of sediment accumulation which are slightly lower than sections to the west. The most significant facies change in the region is the change from red bed deposition at the top of the Kargocha Formation to the brown

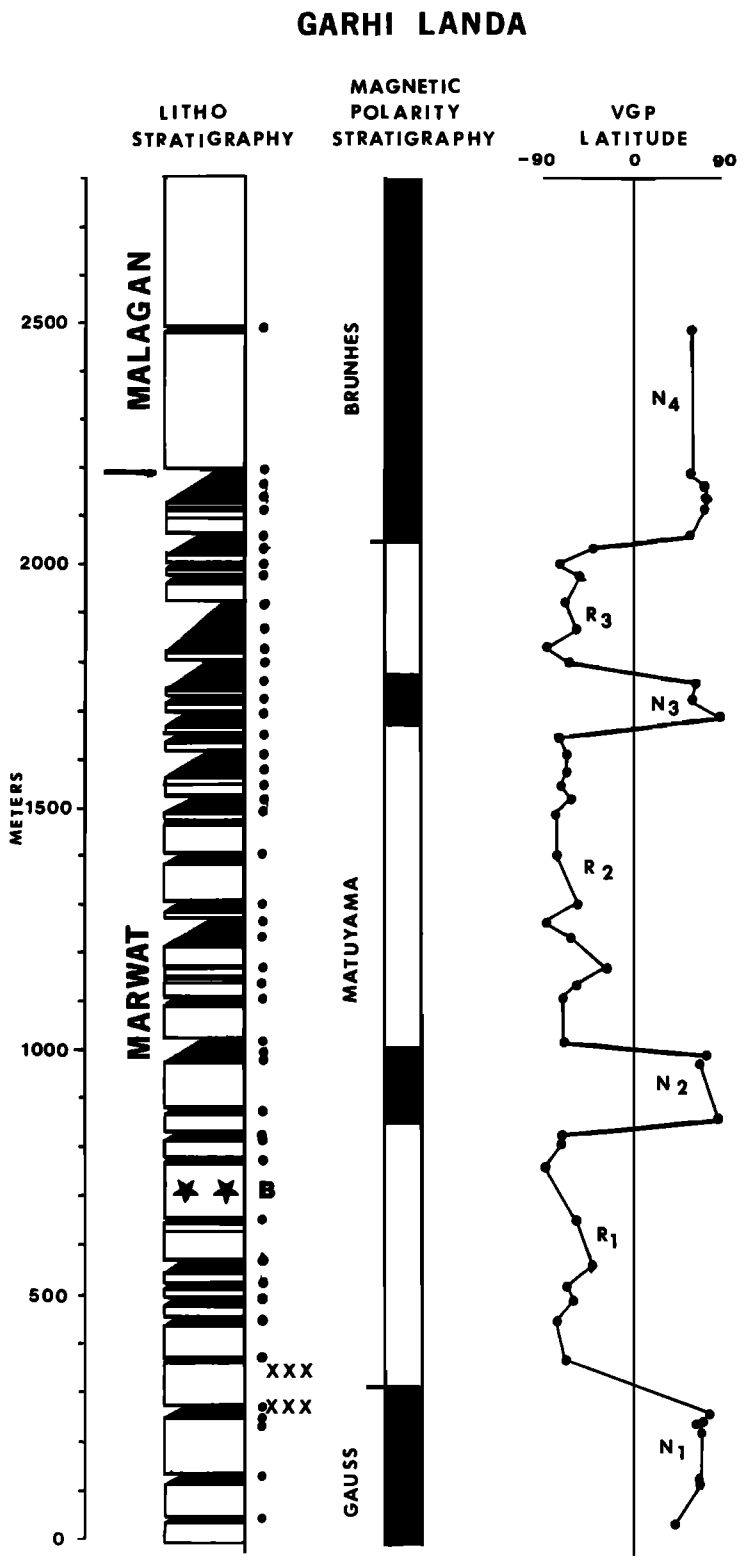



Fig. 8. Garhi Landa. Plot of VGP latitudes against stratigraphic thickness. Plotting conventions are the same as for Figure 6. The letter B and three crosses sign on the right of the lithologic stratigraphic column mark location of the Bain Boulder Bed and the ash units, respectively.

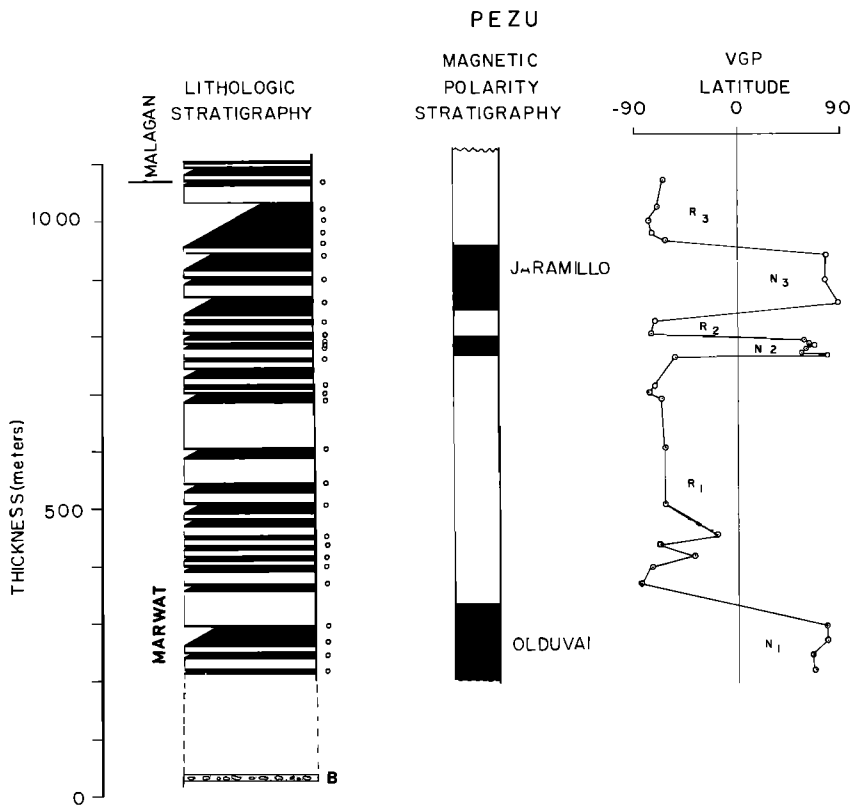



Fig. 9. Pezu Pass. Plot of VGP latitudes against stratigraphic thickness. The plotting convention is the same as Figures 6 and 8.

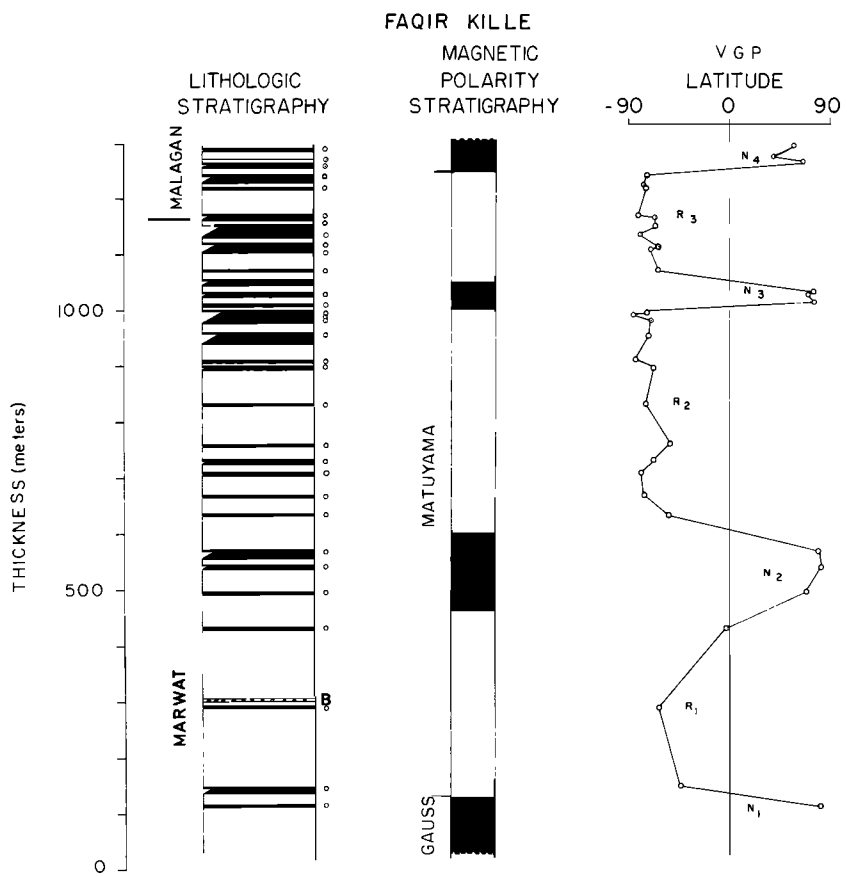



Fig. 10. Faqir Kille. Plot of VGP latitudes versus stratigraphic thickness. Plotting conventions are same as for Figure 6.

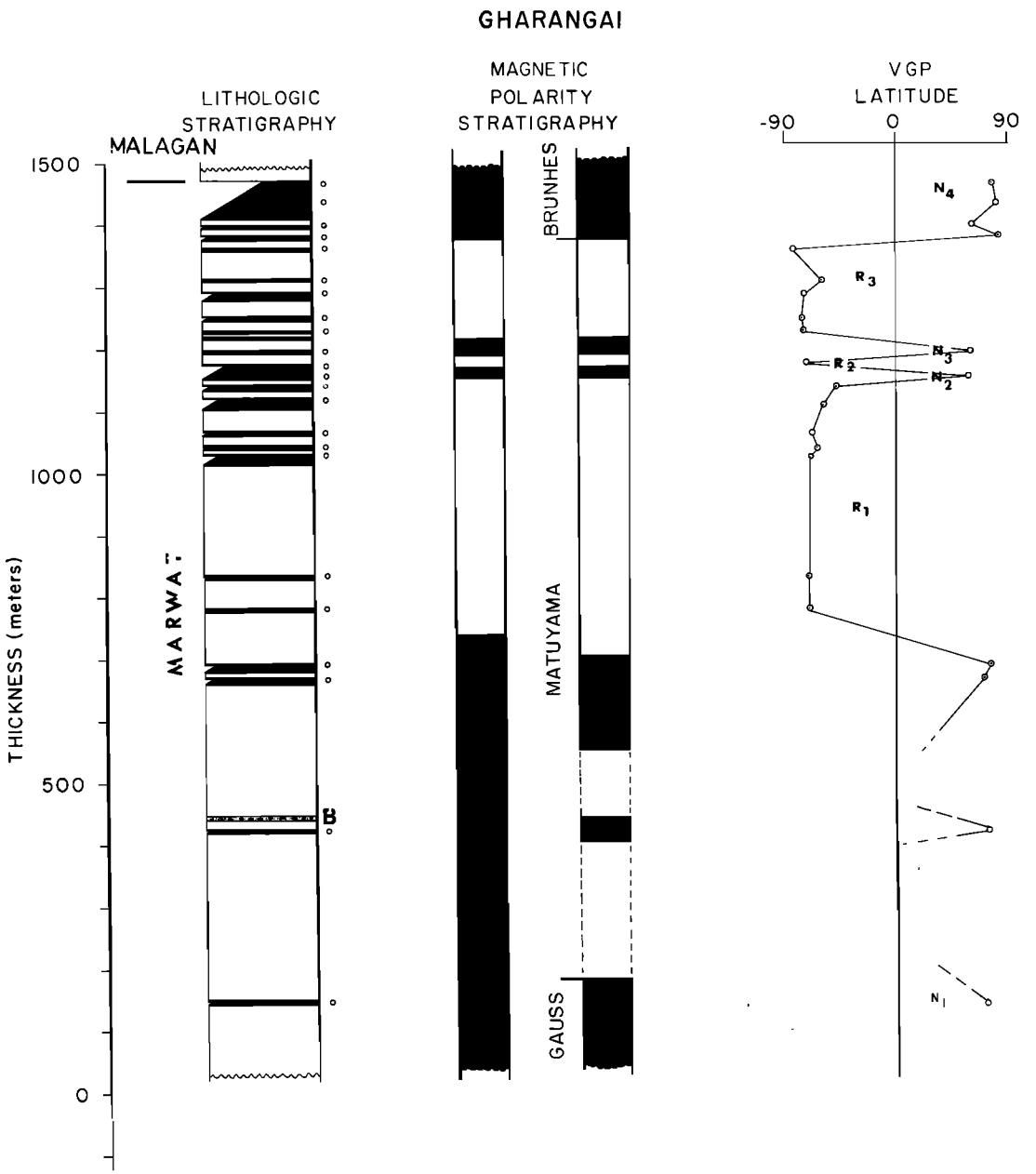



Fig. 11. Gharangai. Plot of VGP latitudes versus stratigraphic thickness. Plotting conventions are the same as for Figure 6. Left column of magnetic polarity stratigraphy is based on the normal convention of plotting a polarity reversal boundary intermediate between two successive interpretations because of the lack of sampling sites in the stratigraphic sequence, marked by dashed lines.

sediments of the Marwat Formation. The top of the Kargocha Formation corresponds to the bottom of the Gauss, and the rate of sediment accumulation for these red beds was calculated using the Cochiti-Gauss interval and yields a rate of sediment accumulation of 60 cm/1000 years, less than the overlying Marwat Formation.

## Plio-Pleistocene Tectonic Movements

The subsidence of the basin containing the upper Siwalik sediments in the Bhittani and Marwat ranges must have begun prior to 3.9 Ma, since sediments of that age are exposed in the core of the Bhittani Range. However, in the Khassor Range, the sediments directly above the Cenozoic unconformity are exposed, and the oldest sediment began to be deposited about 3.19 Ma, just prior to Mammoth subchron. The sedimentation begins with red bed facies (the Kargocha Formation), then in the upper Gauss, the sedimentary sequence becomes dominated by channel sandstones (the Marwat Formation) of the paleo-Indus.

The region continued to subside until some time in the Brunhes chron, when the region began to be uplifted, folded, and faulted. It is probable that this tectonic episode is continuing today. The timing of this orogenic pulse seems to be synchronous with that farther to the east, which produced the Pabbi Hill anticline reported by Keller et al. [1977] and the Rhotas anticline [Opdyke et al., 1979], but later than the folding of the Potwar Plateau, which took place about 2 Ma [Burbank and Raynolds, 1984].

Seeber and Armbruster [1979] have suggested that the Salt Range is the surface expression of motion on a master decollement within the saline series of the Cambrian, which reaches the surface in the Salt Range. It would appear that this motion may have its most

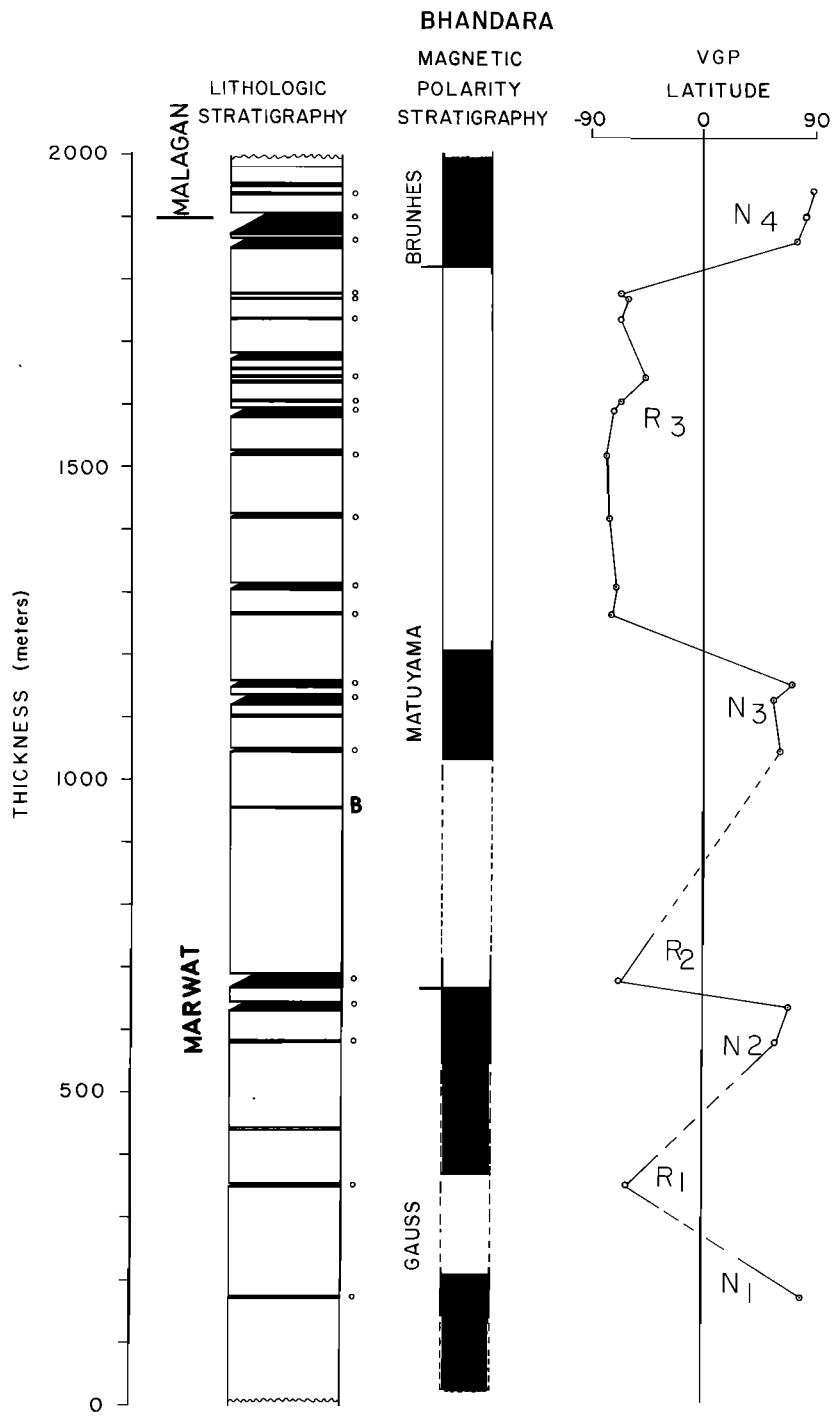



Fig. 12. Bhandara. Plot of VGP latitudes versus stratigraphic thickness. The stratigraphic sequence from 600 m to about 950 m is dominated by thick sandstones and lacks paleomagnetic sites.

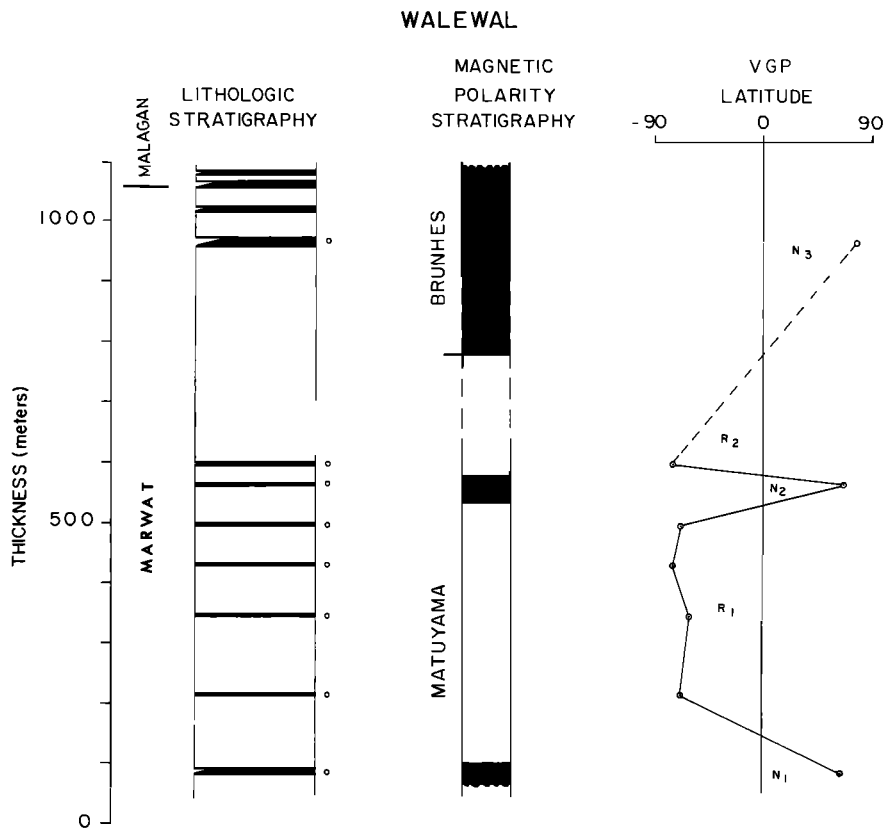



Fig. 13. Walewal. Plot of VGP latitudes versus stratigraphic thickness. Plotting conventions are the same as for Figure 6.

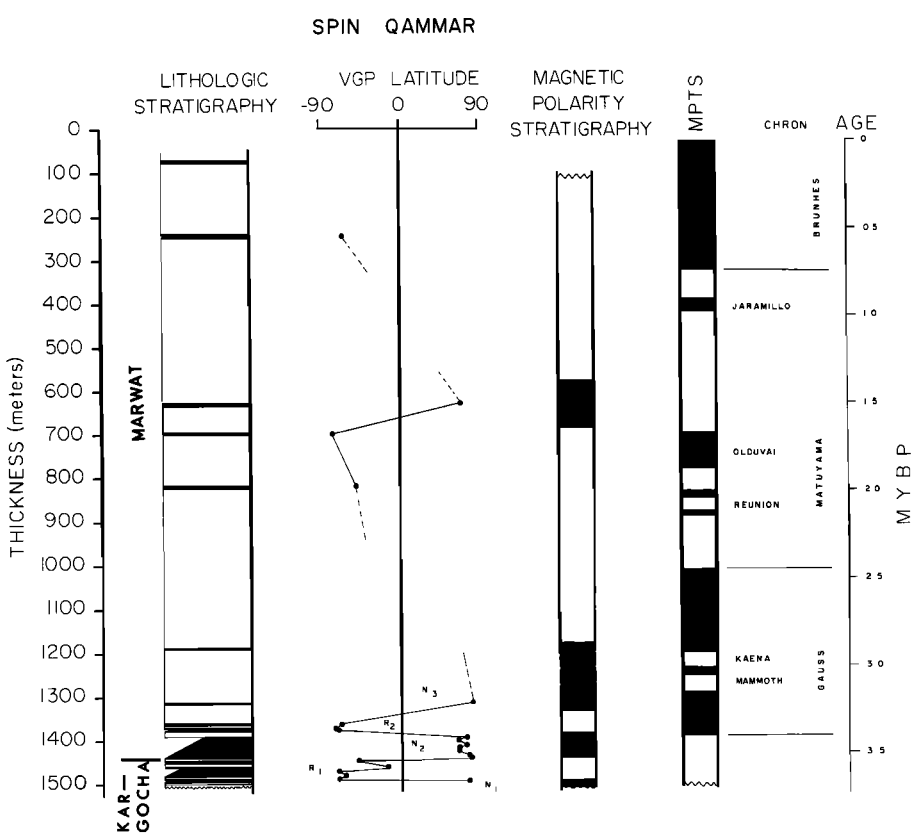



Fig. 14. Spin Qammar. Plot of VGP latitudes versus stratigraphic thickness. Plotting conventions are the same as for Figure 6.

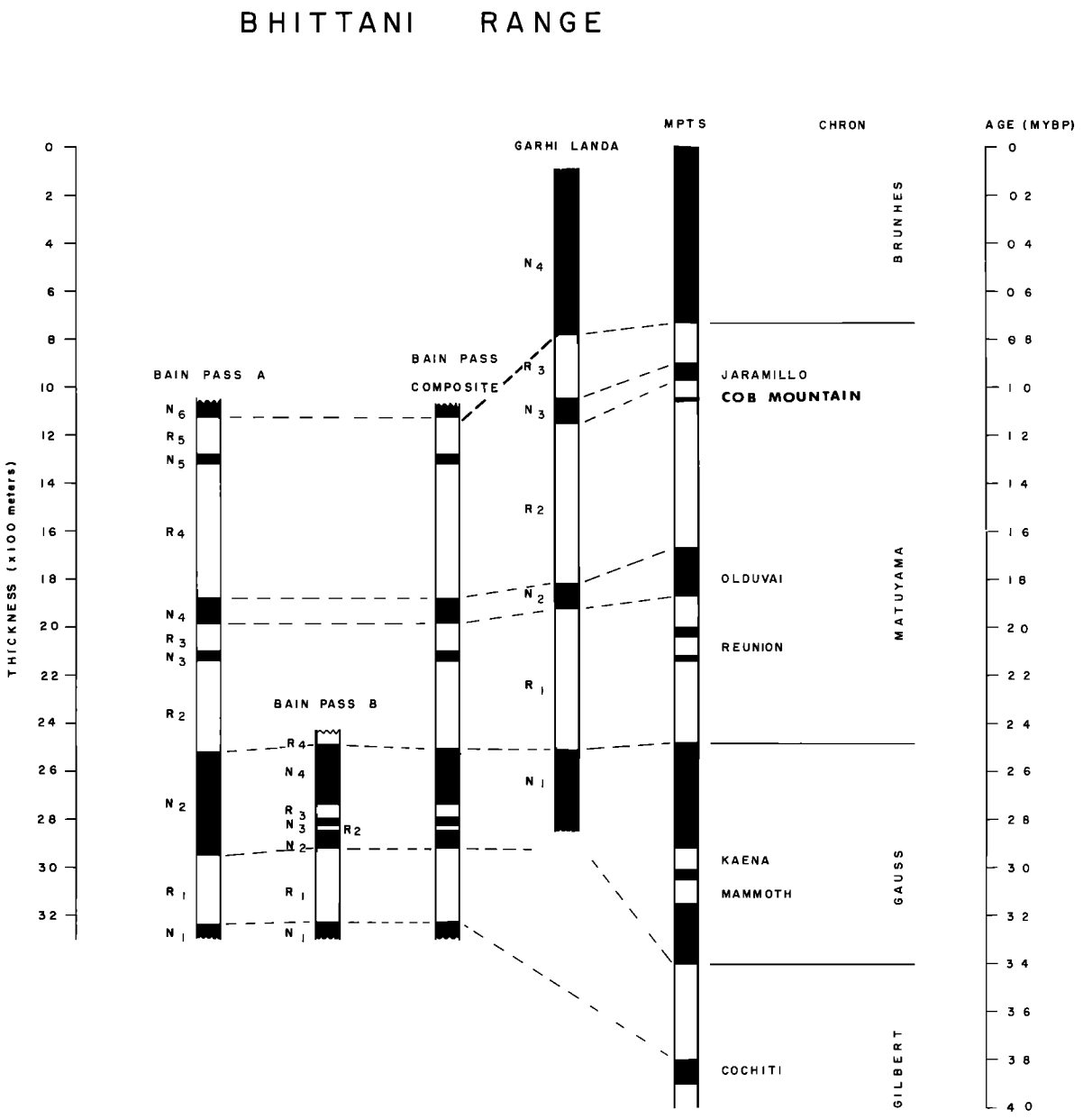



Fig. 15. Correlation chart for the Siwalik Group of the Bhittani Range, based on magnetic polarity stratigraphy. The dashed lines represent proposed correlation with the MPTS.

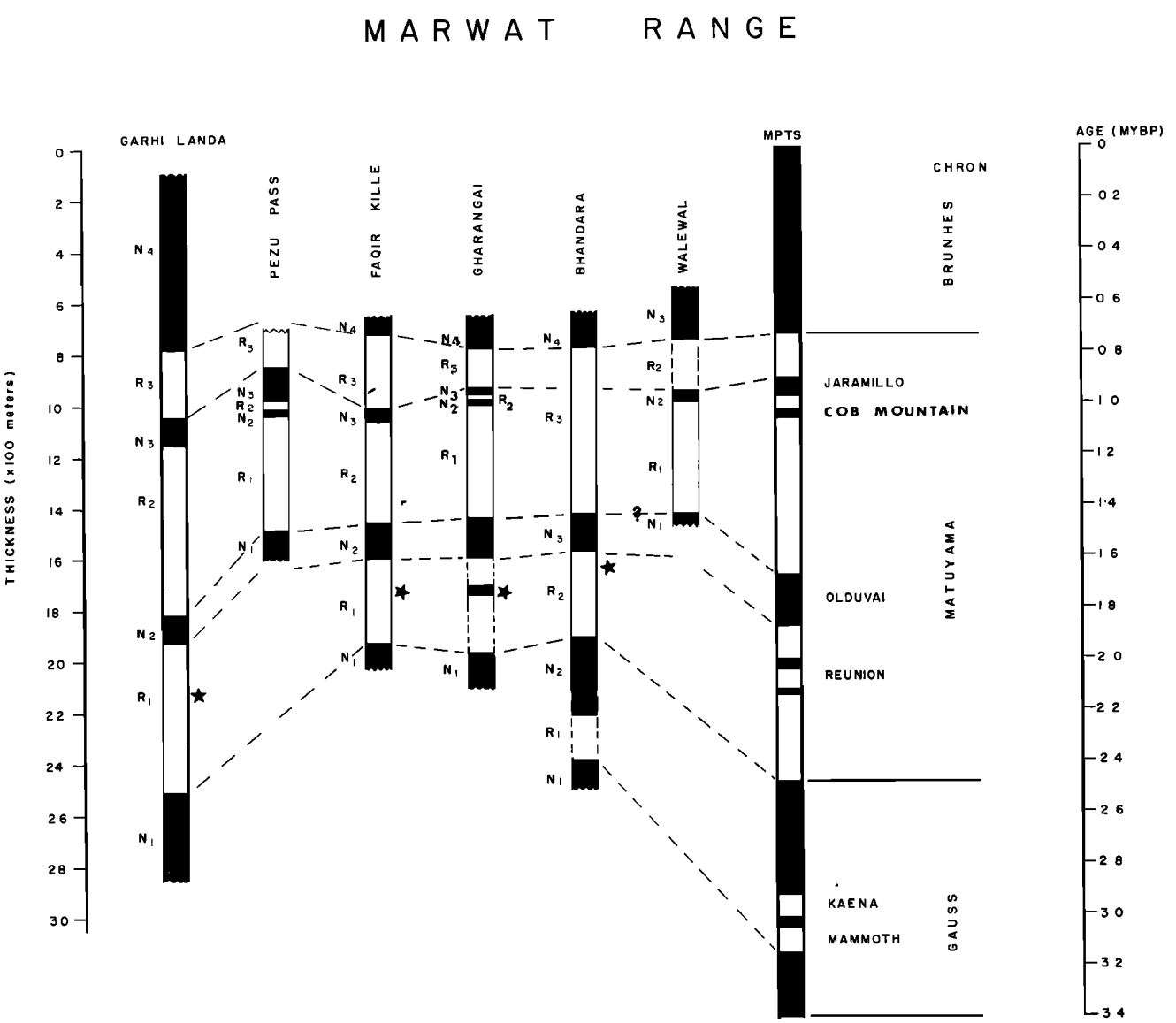



Fig. 16. Correlation chart for the Siwalik Group of the Marwat Range, based on magnetic polarity stratigraphy. The dashed lines represent proposed correlation with the MPTS. The Garhi Landa section is included for correlation of the Bhittani and Marwat ranges. The position of the Bain Boulder Bed is indicated by the stars.

recent expression on both ends of the syntaxis in the Jhelum region, as previously noted, and in the trans-Indus. It is interesting to note that in the Khasor Range, the entire Siwalik sedimentary history of the basin lasts only about 3 m.y. from initiation of subsidence at 3.2 to 4 Ma to uplift and erosion about 0.6 Ma.

In previous studies of the Potwar Plateau region, structure rotations about a vertical axis of varying amounts have been documented [Opdyke et al., 1982; Burbank and Raynolds, 1984]. The later study reported rotations of the Potwar Plateau up to 40° in a counterclockwise direction, with 20° of these rotations occurring in the Brunhes chron. In the study reported here, vertical axis rotation appears to be negligible. Taken at face value, the maximum rotation allowed by the data is a counterclockwise rotation of 5°.

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